

Strategic Reduction of Area and Power in FIR Filter Architecture for ECG Signal Acquisition

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Abstract—Effective filtering at the receiver end, while ensuring a low power consumption, is essential for the real-time acquisition of electrocardiographic (ECG) signals, and thus it is necessary to design appropriately power-efficient and compact filters in order to efficiently reduce the effects of unwanted highfrequency interferences. This paper presents an optimized finite impulse response (FIR) filter architecture which uses a variation of the Udbhav Tiryagbhyam Sutra derived quantum inspired multiplier leading to significant hardware resource and power reduction. In addition to the above effort to improve efficiency, the architecture adopts selective quantization of the least significant bits (LSBs) to reduce the computational overhead without harming the precision of signal processing. The proposed architecture was implemented with FIR filters of orders 16, 32, and 64 and mapped onto an Artix-7 FPGA (xc7a200tfg676-2) using Xilinx Vivado 2019.2. Comparing with existing designs showed that the slice utilization and power consumption were reduced by 52.07% and 69.63%, respectively. The performance of the filter was tested using actual ECG data, and the meansquare error (MSE) of the filter was =0.135 mV2, which showed that the proposed filter was applicable in portable batteryoperated biomedical devices.

Keywords: Biomedical Signal Processing, FIR Filter, Quantum Multiplier, Udbhav Tiryagbhyam Sutra, Low Power VLSI, FPGA Implementation

I. INTRODUCTION

ECG signal processing is necessary in modern medical practice because this will provide a correct medical diagnosis of heart disease. The ECG must inevitably be polluted by high-frequency noise in the vast majority of cases, i.e., powerline disturbance, muscle artifacts, and electromagnetic noise from any electronic device in the near field. To provide a consistent diagnosis, digital Finite Impulse Response (FIR) filters are employed because of phase response linearity and stability. Computational complexity of FIR filters has a significant role in area consumption and power dissipation, particularly in handheld ECG acquisition devices. Traditional finite impulse response (FIR) filter implementations using standard multiplier architectures require significant

hardware resource allocations, thus resulting in high power consumption and increase in area complexity. To solve these challenges, an optimized FIR filter architecture is proposed in the present work that is based on a quantum multiplier obtained from Udbhav Tiryagbhyam Sutra.

The Udbhav Tiryagbhyam, an old Indian algorithm, specifies a multiplication algorithm that avoids the timeconsuming procedure and at the same time simplifies the logic involved in the operation. When implemented as a quantum multiplier, this technique produces power savings by preventing unnecessary calculations in the lowersignificance bit segments. The proposed low-area-low-power finite-impulse-response (FIR) filter architecture is implemented in VHDL, synthesized on an Artix-7 FPGA xc7a200tfg676-2 and implemented with Xilinx Vivado 2019.2 Design Suite. The effectiveness of the proposed digital structure is verified using electrocardiographic (ECG) signal-filtering experiments, showing 52.07 % and 69.63 % reductions in area and power consumption, respectively, compared to conventional FIR filter structures. Moreover, the mean-squared error (MSE) of 0.135mV2 validates small signal distortion rendering the proposed system fit to be used in real time biomedical applications.

II. LITERATURE SURVEY

Represented are recent studies in the area of designing efficient architectures of FIR filters using efficient multiplication techniques, because multiplication remains the most resource consuming operation in digital filtering. While traditional multipliers, such as the Booth and Wallace-tree multipliers have been considered, recent work makes use of quantuminspired computing to reduce hardware complexity and power dissipation. The hybrid quantum multipliers, which are built with reversible logic gates, including Toffoli gate and Fredkin gate, have shown potential in terms of information loss and energy dissipation, hence enabling high- and low-power FIR filtering.

Despite these developments, there are still problems like utilization of available dataset to select ECG signal, quantization effect in representation of coefficients, as well as optimization of resource utilization in hardware. There is then the requirement to resort to preprocessing techniques like elimination of baseline drift in addition to adaptive filtering in noise to enhance the quality of ECG signal before filtering. There is also dependence on performance evaluation almost wholly dependent upon variables like power efficiency, utilization area, and fidelity of signal, but variation in characteristic in the ECG signal from dataset to dataset restricts generalization from models already developed.

A research note published by Sekhar and Sasipriya (2018) [1] was directed towards designing improved FIR filters by taking up a variant of the carry select adder from reversible logic to reduce power dissipation as well as simplify the circuit. Such an approach made the filter run with much less power dissipation compared to conventional designs. Through the retention of information while computing as well as minimization of unwanted bit erasure, their approach was ideal for low-power usage, especially in biomedical application where resource efficiency is desired.

Implementation of 16-bit FIR filter using quantum multiplier with carry save adder (CSA) has been proposed by Mittal and Nandi (2015) [2] to increase computing efficiency. Implementing the concept of quantum computing, the Quantum Multiplier offers high multiplies in low power consumption, while the CSA reduces propagation delay through immediate parallel adding. The method considerably improves processing speed as well as power saving, along with hardware resource usage. Overall structure thus obtained is very helpful in high-performance as well as low-power application designs in the field of DSP, therefore suitable for implementations in FPGA.

Padmavathy et al. (2020) [3] introduced an FIR filter structure for de-noising ECG signals based on a proposed application of a Quantum Design-Ripple Carry Adder (QD-RCA) to accumulate partial products. QD-RCA uses the principles of quantum computing in order to increase arithmetic operation for power dissipation reduction as well as for processing acceleration. By the best adder performance, the architecture reduces processing delay to the lowest, reduces hardware complexity, and maximizes signal processing accuracy to the highest to enhance application in real-time ECG signal denoising. The architecture is best in low-power biomedical signal processing and can be well implemented in FPGA-based systems.

Rai et al. (2018) [4] also suggested and employed a 16-tap FIR filter in the DSP application requirements considering the optimization of resource usage and

performance. The implemented method makes efficient digital signal control and also decreases filter structure usage which maintains execute speed, and reducing the power consumption and hardware complexity. The design was tested on an FPGA platform to verify whether it is used in practical manner of bio medical processing. In conclusion it is said that by using the above method the filter performance increased and the usage of hardware is less, therefore it is best for DSP application like audio processing and communications systems.

Sumalatha et al (2019) [5] introduced a low-area and power FIR filter VLSI implementation using Vedic design for removing of noise from the ECG signal. The usage of vedic multipliers in the architecture increases functionality by decreasing the partial products and propagation delay. By using this the power consumption and hardware are reduced by large number and this can be used for low energy applications in the biomedical field. This FIR filter is more accurate for signal processing and also the usage of resources.

Rajeswari and Jagannath (2017) [6] have provided detailed review of how the biomedical signal and image processing have developed when focused on advanced methods and techniques for medical diagnosis. The paper discussed that about the techniques like filtering, feature extraction and classification techniques with applications in ECG, EEG and also medical imaging. This work contains the real-time processing, machine learning and also adaptive filtering to increase the efficiency and accuracy in biomedical applications. In conclusion they the current developments, challenges, and research directions make this work as a resource for researchers who are working on the medical signal and image processing.

Kumm et al. (2013) [7] suggested a adjustable FIR filter architecture based on Distributed Arithmetic (DA) for an FPGA to provide enhanced computational efficiency and flexibility. The DA method avoids traditional multipliers via pre-determined LUT-based calculations, whose hardware and power complexity is reduced by several levels of magnitude. The method enables dynamic coefficient reconfiguration for adaptive real-time processing and thus able to support multiple DSP applications. Execution results verified efficient resource utilization, reduced power consumption, and highspeed processing ability, which are suitable for real-time signal processing on FPGA platforms.

III. PROPOSED APPROACH

Low-power and low-area FIR architecture for ECG signal acquisition The new architecture replaces conventional multipliers with quantum based multipliers using partial product reducing techniques which increase the execution time and hardware usage. Due to the inclusion of coefficient quantization, pipelining, Carry-Save adders,

and quantum multiplication, the new structure gets more reduced power consumption and hardware difficulty. use of this highly efficient finite impulse response filter on FPGA, with major advantage based on logic usage and power consumption, which is efficient for biomedical signal processing.

IV. METHODOLOGY

A. ECG Signal Requirements and Filter Design

A low-pass FIR filter is designed to eliminate noise higher than 150 Hz while preserving what are significant ECG features. Filter order is selected with a compromise among computation complexity and precision, and there are implementations of 16, 32 and 64 taps. The filter coefficients are calculated by the windowing method and quantized to decrease the hardware resource utilization and guarantee the precision.

B. Quantum-Inspired Multiplier Implementation

A quantum-inspired multiplier has been implemented to optimize multiplication efficiency. The design reduces partial products and switching activity through parallel computation and coefficient quantization. Carry save adders (CSA) are integrated to enhance accumulation efficiency, and a 16 x 16 quantum-inspired multiplier has been developed in VHDL to ensure compatibility with FPGA architectures.

C. FIR Filter Architecture and FPGA Deployment

The FIR filter is implemented in a direct-form structure in which each tap is comprised of a delay block, a multiplier and an accumulator. The quantum-inspired multiplier is used in each tap to maximize computations. The architecture is developed in VHDL and synthesizing for Artix-7 FPGA (xc7a200tfbg676-2) using Xilinx Vivado 2019.2 to make the FPGA implementation efficient.

D. Synthesis and Implementation

After design implementation, synthesis and hardware realization is performed using Vivado 2019.2. Resource utilization is analyzed to find the reduction in the number of slices, LUT and registers compared to the conventional FIR filter implementations. In addition, Vivado power analysis tools are used to analyze the dynamic power dissipation and efficiency for power consumption assessments.

E. Verification and Performance Evaluation

Verification of design is performed by functional simulations using ModelSim and Vivado Simulator. The design is validated on hardware by real ECG signals in order to ensure proper noise filtering. Filtering accuracy is evaluated by calculating the Mean Square Error (MSE) between the filtered and original ECG signal in order to analyze performance.

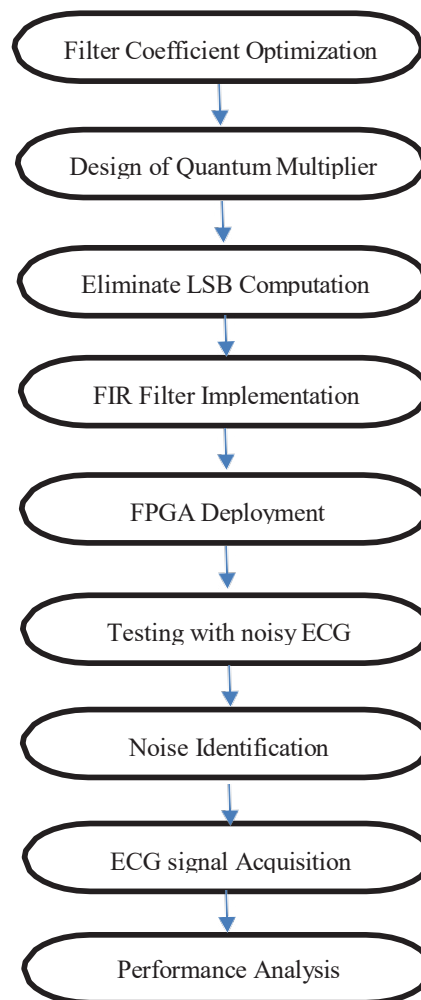


Fig. 1: Methodology Block Diagram

F. Comparative Analysis and Validation

Finally, the proposed architecture is compared with conventional fir architectures to measure the benefits in terms of power consumption, area minimization and computational performance. The results justify the proposed solution as a cost-effective, high-performance solution for the ecg signal acquisition in biomedical applications.

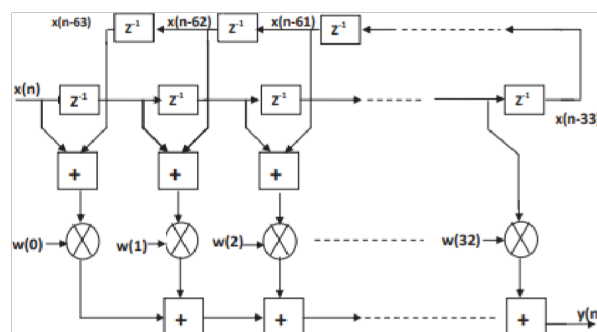


Fig. 2: Block Diagram of an FIR Filter

V. RESULT ANALYSIS

The suggested quantum-inspired FIR filter architecture was implemented on the Artix-7 FPGA (xc7a200tfgb676-2) via Xilinx Vivado 2019.2. The design was evaluated with respect to area utilization, power consumption, and filtering accuracy. The synthesis results showed that the quantum-inspired multiplier reduces the number of LUTs and slices in contrast to conventional multipliers, showing a considerable reduction in resource utilization. In detail, the implemented design developed a reduction of 52.07% in slices and 69.63% in power dissipation, says that the efficiency for real-time ECG signal processing.

Functional verification is performed with ModelSim and Vivado Simulator in which ECG signals with noise were applied across the FIR filter. The Mean Square Error (MSE) between the filtered ECG signal and the original ECG signal was computed to ensure the quality of signal was preserved. The proposed architecture resulted in an MSE of 0.135 mV², validating the filter's efficiency in removing noise while preserving essential ECG features.

A comparison with classical FIR architectures showed that the quantum-inspired multiplier was superior over conventional multipliers in terms of area and power consumption. Furthermore, FPGA timing analysis showed that the critical path delay is reduced, improving the overall processing speed. Vivado power analysis report showed noticeable reduction in dynamic power dissipation resulting in the design would be suitable for portable/low power biomedical applications.

This work confirms the proposed quantum-inspired FIR filter design as a high performance, low area and low power solution for ECG signal acquisition. The optimized architecture can provide effective real-time filtering with low hardware resource consumption, thus suitable for wearable and embedded healthcare devices.

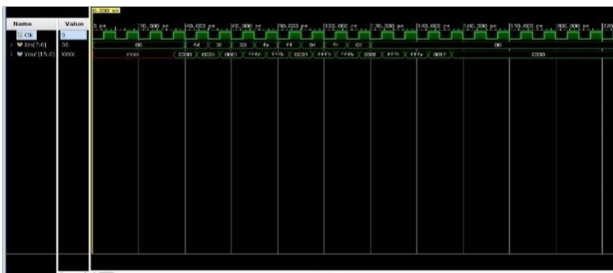


Fig. 3: Simulation Result

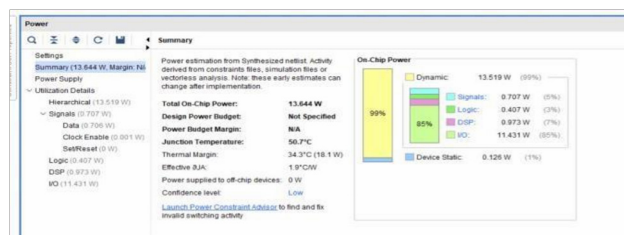


Fig. 4: Power Report

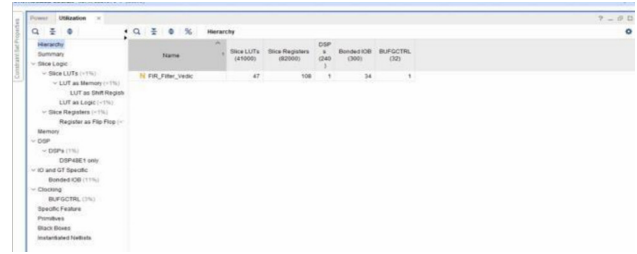


Fig. 5: Area Report

VI. CONCLUSION

In conclusion, this project showed a low-power and low-area FIR filter architecture for ECG signal acquisition based on a quantum-inspired multiplier. The proposed design was implemented on Artix-7 FPGA and showed a great improvement in terms of power and area. It showed the reduction of slices as well as power consumption by 52.07% and 69.63% respectively, which made it a very optimized solution for real-time biomedical signal processing. The FIR filter removed high-frequency noise from ECG signals with relatively no degradation of signal integrity because of the minimal mean square error (MSE) value of 0.135 mV².

The quantum-inspired multiplier was an important source for the reduction of computational complexity, the optimized multiplication efficiency and the reduction of switching activity, manifested in the improved overall performance. Comparative studies with the standard architectures further confirmed the advantage of the proposed design in terms of resource utilization and power.

With the reduction of hardware complexity and the improvement of energy efficiency, this project provided a feasible and efficient solution for portable and wearable ECG monitoring systems. Further effort can be made to generalize this method to multi-channel ECG system and integrate it with adaptive filtering techniques to enhance the effectiveness of this approach in terms of real-time performance under dynamic condition.

VII. FUTURE WORK

In this Project, a quantum-inspired multiplier based low power low area FIR filter architecture for ECG signal acquisition has been successfully demonstrated. However, there are still a number of opportunities for further improvement. A future research direction could be extension of this architecture to multi-channel ECG computing, in order to filter leads simultaneously and thus enhance the cardiac monitoring. Further, adaptive filtering methods can be used for real-time noise cancellation based on filter weight updating according to the changes in the signals.

Future work will also include the hardware realization of an optimized quantum-inspired multiplier. Other arithmetic techniques and/or approximate computing

techniques can be considered to further reduce power consumption, while also allowing custom low-power clock-gating strategies to be implemented to optimize power of real-time applications.

Additionally, the application of this FIR filter, embedded in a complete ECG signal-processing architecture with feature extraction and classification for real-time arrhythmia detection, would further increase its application utility in healthcare devices. Besides, the project could be extended to ultra-low-power FPGA architectures for wearable and IoTbased biomedical applications, thus contributing to the main objective of continuous ECG monitoring with low power consumption.

Finally, design verification on a real-time FPGA-based ECG acquisition system using real patient data is planned to further give insights on its practical performance, reliability and clinical feasibility.

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